

Taewoon Kang

Integrated M.S. and Ph.D. Student
Korea University

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Education

Korea University Department of Computer Science and Engineering Ph.D. Student GPA: 4.35 / 4.5 Advisor: Gunjae Koo	Mar. 2023 – Present Seoul, Korea
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Sangmyung University Department of System Semiconductor Engineering B.S. in System Semiconductor Engineering GPA: 4.28 / 4.5 Advisor: Yongwoo Kim Thesis: <i>An Implementation of CNN-based Real-Time Face Mask Detector using Jetson Nano</i>	Mar. 2019 – Aug. 2022 Cheonan, Korea
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Research Interests

AI accelerators; HW/SW co-design for SoCs; processing for AI (NPU);
computer architecture; processing-in-memory.

Experience

University of Southern California Visiting Scholar Ming Hsieh Department of Electrical and Computer Engineering Supervisor: Prof. Murali Annavam	Jan. 2026 – Present United States
SK Telecom Intern Developed spam detection methods using AI.	Jun. 2023 – Aug. 2023 Korea

Teaching Experience

Teaching Assistant Basic of software Digital logic design	Spring 2025 Spring 2023, Fall 2023
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Academic Service

Artifact Evaluation Committee IEEE/ACM International Symposium on Microarchitecture (MICRO)	2026
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Publications | International

1. Narae Kim, Wonjae Choi, **Taewoon Kang**, Yujin Kim, Seongpil Yang, Myung Kuk Yoon, Yunho Oh, Gunjae Koo. "Fine-Grained DRAM Refresh Exploiting Divergent Bit-Error Tolerance in AI Workloads." [Under review]
2. **Taewoon Kang**, Jonghyun Lee, Yujin Kim, Murali Annavaram, Gunjae Koo. "LogPIM: Low-Overhead Logarithmic Arithmetic for HBM-Based Processing-in-Memory." [Under review]
3. Jong Hyun Jeong, Yuan Feng, **Taewoon Kang**, Gunjae Koo, Hyeran Jeon. "GPO: Orchestrating GPU-PIM Computing for Sparse-Dense Workload Acceleration." [Under review]
4. Jonghyun Lee, **Taewoon Kang**, Pratheek B, Yongqin Wang, Rachit Rajat, Mengyuan Li, Gunjae Koo, Daniel Wong, Murali Annavaram. "FLASH: Fault-Led, Attested, Secure Hybrid-Key Checkpoint for Confidential GPU." *Proceedings of the 2026 ACM Annual Technical Conference*, 2026. [To appear]
5. **Taewoon Kang**, Namhun Kim, Jihun Lee, Gunjae Koo. "SoftmaxPIM: An HBM-Based PIM Architecture for Accelerating GeMV-Softmax Execution Pipeline." *IEEE Computer Architecture Letters*, 2026.
6. **Taewoon Kang**, Namhun Kim, Geonwoo Choi, Taeweon Suh, Gunjae Koo. "SparsePIM+: Accelerating SpMV on HBM-Based PIM via Logic-Die Accumulators with Opportunistic TSV Utilization." *Journal of Systems Architecture*, 2026.
7. Sunchae Kim, **Taewoon Kang**, Sangwon Shin, Taeweon Suh, Yibin Yang, Gunjae Koo. "Sumcheck-PIM: An Efficient HBM-Based PIM Architecture for Linear Complexity Zero Knowledge Proofs." *ACM International Conference on Supercomputing*, 2026.
8. **Taewoon Kang**, Geonwoo Choi, Taeweon Suh, Gunjae Koo. "SparsePIM: An Efficient HBM-Based PIM Architecture for Sparse Matrix-Vector Multiplications." *ICS '25: Proceedings of the 39th ACM International Conference on Supercomputing*, 2025.

Publications | Domestic

1. Seongpil Yang, **Taewoon Kang**. "Performance Analysis of a Per-Row Activation Counting Mechanism in DRAM." *The Institute of Electronics and Information Engineers, 2025 Summer Conference*, 2025.
2. Sang-un Jo, **Tae-woon Kang**, Yongwoo Kim. "A Design and Implementation of MP3 Player using ARM Cortex-M0 DesignStart." *The Korean Institute of Electrical Engineers Workshop*, Yeosu, Korea, Jul. 2022.
3. Sang-un Jo, **Tae-Woon Kang**, Yongwoo Kim. "A design and verification of ARM Cortex-M0 DesignStart MCU system using open source IP." *ICS 2022*, Jeonju, Korea, Apr. 2022.
4. **Taewoon Kang**, Yongwoo Kim. "An Implementation of CNN-based Real-Time Face Mask Detector using Jetson Nano." *JKAIS*, Vol. 23, 2022.

Patents | Domestic

Gunjae Koo, Taewoon Kang. “Apparatus and Method for Sparse Matrix-Vector Multiplication Based on Processing-In-Memory”

KR-Registration No. 10-3008421

Awards and Honors

All awards below: Korea University, Korea.

Support Program for Outstanding Graduate Students’ International Joint Training NRF (Brain Korea 21), Korea	Jan. 2026
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All awards below: Sangmyung University, Korea.

The top prize of CISC Career day	May 2022
Header (in one’s dept.) Scholarship	Mar. 2022
Header (in one’s dept.) Scholarship	Sep. 2021
“BONGSA” Scholarship (Leader of Department)	Sep. 2021
Secondary Header (in one’s dept.) Scholarship	Mar. 2021
Work Scholarship	Mar. 2021
“BONGSA” Scholarship (Leader of Department)	Mar. 2021
Secondary Header (in one’s dept.) Scholarship	Sep. 2020

Coursework

- Parallel processing; AI accelerator architecture
- Advanced computer architecture; storage system architecture
- ARM Cortex-M series IP implementation
- Electrical circuit; microelectronic circuit
- Semi-custom ASIC design with Synopsys design tool; VLSI design
- Digital logic circuit; computer architecture and design; design with Verilog; SoC design
- Semiconductor physics; semiconductor fabrication process
- Calculus; engineering mathematics

Skills and Techniques

- C, C++, Python
- GEM5, DRAMsim3
- Chip design verification with Xilinx/Altera FPGA
- Python (PyTorch); C language (Atmel, Keil)
- Verilog 2001; SystemVerilog
- ARM Cortex-M0/M3 implementation
- RISC-V simple processor design
- Semi-custom ASIC design with Synopsys EDA tool